

Testing the Reader on Real Devices

How to check the reader improvements on phones, tablets and laptops — three tiers, a device matrix, a walkthrough and the traps

Reader batches 1-2 · September 2026

Three tiers, cheapest first

Most reader bugs are layout and pointer-logic bugs, and emulation finds those in minutes. A smaller set are physics and platform bugs — scroll feel, safe areas, Safari — and only a real device finds those. A last few only show on production, after a deploy. Spend time in that order.

1 Browser emulation for layout and pointer logic

Minutes per pass, and it catches most things. Chrome DevTools' device toolbar at 375×812 (phone), 768×1024 and 1024×768 (tablet, both ways) and 1440 (laptop). Emulation flips `pointer: coarse`, which is what gates the 44px scrubber, the swipe and tap page turns and the tablet-default width — so it exercises the real code paths, not a lookalike.

CHECK Page-mode tap zones · the Aa panel and its live preview · margins and extra-wide · the progress hairline · the up-next card · the return pill and the cross-device offer · Undo toasts.

2 Real devices on the same Wi-Fi, against the dev server

The only way to feel scroll physics, safe-area insets and Safari's quirks. Run the frontend with `--host`, open `http://<mac-ip>:5180` on the phone. The API must list the phone's origin in `CORS_ALLOWED_ORIGINS` and `PUBLIC_API_BASE_URL` must point at the Mac's IP, not localhost. Test in Safari *and* in the installed PWA (home-screen icon).

CHECK The service-worker cache is where "feature absent" false alarms live: unregister it and clear site data before trusting a negative.

3 Production spot-checks after every deploy

Per the deploy playbook's traps: book URLs with the trailing slash (the no-slash form is a 5.7 KB SPA shell that 200s forever), and confirm the web build actually changed — compare the JS bundle hashes — before hunting for a feature. The API build lags frontend-only PRs; an endpoint exists when it answers 401, not 404.

CHECK Two signed-in devices on production are the only honest test of anything that syncs.

THE RULE

Never call a feature absent until the service worker is unregistered, site data is cleared, and the bundle hashes prove the build you are looking at is the one you shipped.

A device matrix that covers the risk

Four devices, each on the list for something the others cannot show.

DEVICE	WHAT IT ALONE REVEALS
iPhone · Safari + installed PWA	Safe-area insets, 100vh, rubber-band scroll, the toast stack and the return pill sitting above the footer, text selection vs the tap-to-turn zones.
Android · Chrome (a mid-range phone)	Coarse pointer, low-end performance of the idle prefetch and page-mode layout, the 44px scrubber, back-gesture vs swipe-to-turn.
iPad · portrait and landscape	The 768-1023 tablet-default width, two-thumb tap zones, rotation re-layout, page mode with a hardware keyboard attached.
Laptop · 1280 and a scaled 4K display	Extra-wide measure, keyboard navigation and focus rings, the Aa panel's 70vh cap, hover states on the pills.

Setup recipes

Copy-paste; the two env files must agree on the Mac's LAN address.

FRONTEND ON THE LAN

```
cd frontend && npm run dev -- --host --port 5180 --strictPort
# then on the phone: http://<mac-ip>:5180/books/<slug>/1/
```

API REACHABLE FROM THE PHONE

```
# backend/.env
CORS_ALLOWED_ORIGINS=http://localhost:5180,http://<mac-ip>:5180
# frontend/.env
PUBLIC_API_BASE_URL=http://<mac-ip>:8000
```

CLEAN SLATE BEFORE TRUSTING A NEGATIVE

```
# DevTools → Application → Service Workers → Unregister; Storage → Clear site data
# or in the console:
navigator.serviceWorker.getRegistrations().then(rs => rs.forEach(r => r.unregister()));
caches.keys().then(ks => ks.forEach(k => caches.delete(k)));
```

DID THE WEB BUILD CHANGE?

```
curl -s https://ochorus.com/books/<slug>/1/ | grep -oE '_app/immutable/[^\x27]+\.js' | sort | md5
```

The walkthrough, on each device

In this order, once signed out and once signed in — sync and the toast stack behave differently. Tick as you go.

Open cold

- ☐ Open a mid-book chapter from the home screen / a deep link: the position restores to the paragraph, not the top; no progress hairline in page mode, hairline present in scroll mode.
- ☐ Signed in, with the other device further along: the “further along on another device” pill appears within a second; **Continue** · **ch. N** lands on the paragraph; no “back to where you were” pill follows it; dismiss holds for the book.

Type & layout

- ☐ Open Aa: the preview line follows size, typeface, width and margins live; Generous margins move the chrome bar with the text; extra-wide is offered on laptops; Margins are hidden in page mode.
- ☐ First run on an iPad in portrait defaults to the wide measure; a stored choice beats it.

Chapter end

- ☐ The Next card shows the next chapter’s reading time and its opening line without the button growing under your thumb; a Contents link sits below it mid-book.
- ☐ Reading times everywhere say “· at your pace” in the footer after ~7 minutes of reading and shrink or grow to match; listening and scrubbing do not change them.

Turn pages

- ☐ Tap right/left thirds and swipe both ways in page mode; RTL book flips both; rubber-band at the chapter ends; a fast flick turns one page, not two.
- ☐ Scroll mode: the footer scrubber has a ≥44px thumb target on touch; the last line of prose clears the footer; “Page N / M” updates as you go.

Marks & bookmarks

- ☐ Highlight, recolour, then un-highlight: “Removed · Undo” appears; Undo restores colour and note; wait 6 s and it goes away.
- ☐ Remove a bookmark in the contents drawer: the Undo is *inside* the drawer (reachable by Tab); close the drawer and it moves to the corner toast.

Modes & themes

- ☐ Focus mode on a phone: the hairline stays, a screen reader hears “Page 4 / 9” as pages change; Escape or the summon gesture brings the chrome back.
- ☐ Dark theme and the sepia palette: pills, toasts and the drawer stay legible; the accent passes contrast on both grounds.

The two-device pass

Same account on both. A and B can be two browser profiles on one Mac for a first pass; the real thing is a phone and a laptop.

1. On **A**, open a book and read to chapter 7; scroll a few paragraphs in. Wait a second for the push.
2. On **B**, having last read chapter 3 of the same book, open chapter 3 from the book page. Within a second the pill says *You’re further along on another device* · *Continue* · *ch. 7*. Take it: you land on A’s paragraph and no “back to where you were” pill follows.
3. Turn a chapter on B: no second pill. Dismiss holds for the book until the page is reloaded.
4. On **B**, reopen chapter 3 without scrolling, then check A: A’s position on the server is unchanged — a bare open no longer counts as reading. Scroll on B, and it does.
5. Bookmark a paragraph on A; open the contents drawer on B: it is there. Remove it on B, Undo, check A still has it after a reload.
6. Read for ten minutes on A: A’s footer says · *at your pace*; B’s does not — pace is per device.

Traps

Each of these has cost a session at least once.

- EMULATION CANNOT TELL YOU** Whether iOS text selection fights the tap-to-turn zones, and whether the toast stack overlaps the audio player on a small phone. Both need real hardware.
- HIDDEN PREVIEW PANES HAVE NO VIEWPORT** A hidden in-app browser pane reports `innerHeight: 0` and `visibilityState: hidden`: nothing scrolls, and visibility-gated features correctly do nothing. Emulate a device size before calling a feature broken.
- MODULE INSTANCES AFTER AN EDIT** Overriding a store from the console works only until the file is edited; after HMR the app holds a timestamped module URL and your import is an inert twin.
- THE LOCALIZED-PAGE TRAP** A stored en preference de-localizes `/lg` URLs in a persistent session; wipe storage or use curl for ground truth.
- TWO-DEVICE SYNC** Bookmark sync and the cross-device offer only prove themselves with two signed-in devices on the same account; one browser with two profiles is the cheapest stand-in.

Sign-off

A release is tested when every row has a date and initials.

DEVICE	PASSES	DATE	BY
Phone · Safari	Open cold · Turn pages · Marks · Chapter end · Focus		
Phone · PWA	Open cold · Offline reopen · Toasts vs player		
Android · Chrome	Turn pages · Scrubber target · Back gesture		
iPad · both orientations	Tablet default width · Two-thumb zones · Rotation		
Laptop	Extra-wide · Keyboard · Aa panel		
Two devices, signed in	Bookmark sync · Cross-device offer · Pace stays per device		

Where the code lives

For the bug report: name the file and the state you were in.

BEHAVIOUR	FILE
Page turns, tap zones, swipe	frontend/src/lib/pageGestures.ts · routes/books/[slug]/[order]/+page.svelte
Aa panel, preview, margins, width	lib/readerPrefs.svelte.ts · lib/components/ReaderControls.svelte
Hairline, up-next card, return pill, sync pill	routes/books/[slug]/[order]/+page.svelte · lib/resumeSync.ts
Undo for highlights / notes / bookmarks	lib/undo.svelte.ts · lib/undoable.ts · lib/components/PwaToasts.svelte
Reading pace and time estimates	lib/pace.ts · lib/readingPace.svelte.ts · lib/reading.ts
Position sync (API)	backend/reading/views.py ProgressView · lib/readingSync.ts · lib/progress.ts